

To: The President

From: Gabriel Lopez

Re: The President must decide whether to authorize federal grants for a phased transition from diesel to electric school buses, prioritizing high-need public school districts

Date: April 20, 2026

1. BLUF Statement (Bottom Line Up Front)

The President should put a call to action on a phased federal grant approach, emphasizing high-need and lower income public schools to raise funds towards electric school buses. This decision is urgent. A statistic shows that more than 21 million American children ride diesel school buses daily, which in turn exposes them to the emissions directly linked to asthma, cancer, lung disease, and diminished cognitive development(U.S. Environmental Protection Agency, 2026). The most affected are those in low income and minority communities, because of the location in which diesel is most concentrated, including the lack of options for airborne protection(Popkin, 2024). The EPA's Clean School Bus Program has seen success following an award of around \$3 billion to 1,300 school districts around the country.(U.S. Environmental Protection Agency, 2026). Although, the difference in demand to available funding is far too great, and with the funding freeze in January 2025, the program has uncovered its susceptibility towards political disruption, thus creating a long term legislative decision a necessity. This memo presents the benefits and costs of an increased federal investment, analyzes targeted versus universal funding methods, and suggests a phased approach, centered around equity, as the most efficient and fiscally responsible way into the future.

2. Explanation of the Pros and Cons of the Policy

The EPA's Clean School Bus Program(CSBP), is the federal government's way of handling emissions from diesel school buses. Created by the Bipartisan Infrastructure Law of 2021, seeing around \$5 billion in the span of 5 years into swapping out diesel buses for cleaner alternatives(U.S. Environmental Protection Agency, 2026). In a matter of 3 cycles of funding, almost \$3 billion was designated to 1,300 school districts in every state(U.S. Environmental Protection Agency, 2026). Some may see this as a considerable achievement, others highlight the setbacks it comes with, seen in the funding freeze in January 2025, putting districts in doubt. Since demand is considerably greater than available funding, districts all over the country are in need of support(U.S. Environmental Protection Agency, 2026). These effects are an indication towards the ineffectiveness of the current program, insufficient in handling the scope of the issue.

The argument in favor of increased funding is backed behind research and empirical evidence. A study monitoring 275 elementary school students revealed that switching to cleaner buses led to a 16% decrease in lung inflammation and corresponded with considerably fewer absences each year (Adar et al., 2015). In addition to health benefits, research examining more than 2,600 retrofitted school buses in Georgia revealed that lowering diesel emissions led to significant increases in standardized test scores, an impact similar in scale to decreasing class sizes (Persico & Johnson, 2019). These results demonstrate that funding cleaner buses acts as both a public health measure and an educational strategy with proven, measurable outcomes(Adar et al., 2015; Persico & Johnson, 2019).

The main objection to increased funding revolves around expense. A new electric school bus costs between \$300,000 and \$400,000, 4 times the amount of a diesel bus(Popkin, 2024), all while charging infrastructures may increase expenses by an extra \$140,000 for each site(Popkin, 2024) The World Resources Institute's data shows rural and remote locations face increased obstacles, like restrictions in grid capacity, deficiencies in workforce training, and limited administrative abilities to manage federal grant applications(Worker & Coursar, 2023). These challenges should not halt action, but instead recognize the need for a gradual, centralized strategy rather than a blanket rule that could overwhelm districts that are less capable.

3. Explanation of the Recommendations

Considering the evidence provided, it's suggested that the President approve the expansion and long-term reauthorization of the EPA's CSBP, via a tiered federal grant system that focuses on high need, low income, and rural school districts. This suggestion is favored over two other options: preserving the current situation and adopting a universal, non-targeted funding model that allocates grants uniformly to all districts irrespective of their requirements.

The current situation is not a sustainable choice. As outlined in the previous section, existing funding levels are inherently inadequate, politically precarious, and unable to satisfy the recorded demand from qualifying districts(U.S. Environmental Protection Agency, 2026). Proceeding without increased commitment ensures that the health and educational effects of diesel exposure will endure for millions of children, mainly affecting communities that are the least capable of handling them on their own(Adar et al., 2015; Persico & Johnson, 2019).

A universal funding model, though seemingly fair, carries its own hazards. Allocating grants uniformly to all districts would dilute scarce resources, diminishing the program's measurable effectiveness and channeling funds to affluent districts that likely already have the administrative capability and local funding to achieve electrification themselves(Worker & Coursar, 2023). The World Resources Institute discovered that rural districts with great needs and low-income areas encounter multiple obstacles that static funding models do not resolve (Worker & Coursar, 2023). Resources allocated to areas of high need and impact are not only financially cautious but also ethically reasonable.

The phased, targeted grant model proves to be more effective by concentrating initial investments on districts facing the greatest health challenges, resulting in measurable outcomes, improved health metrics, reduced absenteeism, and academic enhancement that bolster the case for continued funding, while allowing costs to decrease organically as the electric bus market grows(Popkin, 2024). This approach directly addresses the equity concerns noted by the evidence while remaining administratively feasible within the existing CSBP structure(U.S. Environmental Protection Agency, 2026; Worker & Coursar, 2023).

4. Implementation or Next Steps

The phased federal grant program's implementation should be three stages within a six-year timeframe, with each phase extending the results of the prior stage.

During Phase 1 (Years 1–2), the government ought to renew and increase funding for the CSBP with at least \$2 billion in new allocations, focusing on the 500 highest-need districts determined by the EPA's current prioritization standards (U.S. Environmental Protection Agency, 2026). Financing in this stage must encompass the total expenses related to electric buses, charging facilities, and employee training. The Joint Office of Energy and Transportation ought to be responsible for offering direct technical support to districts that lack the administrative resources to manage the federal grant process.

During Phase 2 (Years 3–4), eligibility to include all Title I school districts across the country. Grant amounts ought to be reduced as costs for manufacturing electric buses decrease because of market expansion (Popkin, 2024). Districts that finished Phase 1 transitions must be assessed to produce performance data such as health outcomes, attendance figures, and academic indicators (U.S. Environmental Protection Agency, 2026), that will guide Phase 2 program development and enhance the legislative argument for ongoing funding.

In Phase 3 (Years 5–6), the remaining districts ought to be included in the program via a mix of federal grants and low-interest loans from the federal government. The critical risks on all stages include supply chain restrictions in bus production, utility grid limitations in rural areas, and the risk of disrupted funding (Worker & Coursar, 2023). Although, these can be remedied through long term funding decisions, including the cooperation with the Department of Energy and domestic bus manufacturers to create the supply chain line.

5. Conclusion

The proof is evident and the way ahead is practical. Each school day that goes by without a continued federal commitment to transitioning to electric school buses is another day that countless American children, particularly those from low-income and minority backgrounds, inhale diesel fumes on their journey to school, endure avoidable respiratory issues, and perform poorly academically as a consequence. EPA's CSBP showed that the transition is practical, efficient, and impactful. It lacks the lasting equity focused legislative support needed to support children of need. Should the President take action, the commitment to education and health demonstrates the administration prioritizing the futures of Americans. If the President fails to take action, the current situation will further exacerbate those inequalities for many years ahead. The decision involves both a policy aspect and a moral aspect.

6. Annexes

Annex A: SWOT Analysis–Federal Investment in Electric School Bus Transportation

The SWOT analysis below depicts the strengths and weaknesses of the phased federal grant approach, including the external opportunities and threats which could impact the implementation of electric school buses. The evaluation provides evidence seen in Section 3, displaying the program's benefits and potential to exceed the challenges, specifically when a gradual, strategical implementation is utilized. The primary internal hurdle substantial initial expenses is effectively addressed by the recommendation's phased approach, whereas the largest external opportunity, an expanding electric vehicle market reducing bus costs over time, bolsters the long-term financial argument for taking action now instead of delaying.

STRENGTHS	WEAKNESSES
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• <u>Eliminates harmful diesel emissions</u> • <u>Long-term cost savings (\$7,000+/yr per bus)(Electric School Bus Initiative, 2025)</u> • <u>Improves student health & academic performance</u> • <u>Existing federal program infrastructure (EPA CSBP)</u>	• <u>High upfront cost (\$300K–\$400K per bus)(Popkin, 2024)</u> • <u>Charging infrastructure gaps, especially rural areas</u> • <u>Workforce training requirements</u> • <u>Political resistance to federal spending</u>
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OPPORTUNITIES	THREATS
• <u>Equity-focused funding for underserved districts</u> • <u>Phased implementation reduces fiscal strain</u> • <u>Vehicle-to-grid (V2G) energy benefits</u> • <u>Growing EV market lowering bus costs over time</u>	• <u>Federal funding freezes or program rollbacks</u> • <u>Grid capacity limitations in rural regions</u> • <u>Supply chain delays in bus manufacturing</u> • <u>Inequitable adoption favoring wealthier districts</u>

References

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